

课程重点

1. 软件体系结构的基本概念
2. 软件质量属性相关
3. 软件架构模式
4. 软件架构设计过程
5. 设计模式
6. 软件架构文档化
7. 软件架构评估
8. 软件产品线
9. 微服务架构

Summary 中的点

1. Software Architecture in General

1. What is software architecture? (Structure, Elements, Relationships, Design)
2. What does a software architect do?
3. Where do architectures come from? (NFRs, ASRs, Quality Requirements; Stakeholders, Organisations, Technical Environments...)
4. Architecture Views (Logical view, Process view, Physical view, Development view + Use case scenarios...)
5. Architectural activities and process
6. Software architecture knowledge areas

2. Quality Attributes

1. Software Requirements (Functional requirements, Quality requirements (NFRs), Constraints)
2. Modeling quality attribute scenarios: (Source, Stimulus, Artefact, Environment, Response, Measure)
3. Availability, Interoperability, Modifiability, Performance, Security, Testability, Usability, X-ability
4. Tactics for quality attributes
5. Architecturally Significant Requirements (How to gather and identify ASRs: Requirements, Interviews, Business goals, Utility tree)

3. Architecture Patterns

1. Architecture Patterns (Context, Problem, Solution: elements + relations + constraints)
2. Module Patterns (Layered pattern)
3. Component-Connector Patterns (Broker pattern, Model-view-controller pattern, Pipe-and-filter pattern, Client-server pattern, Peer-to-peer pattern, Service-oriented pattern, Publish-subscribe pattern, Share-data pattern)
4. Allocation Patterns (Map-reduce pattern, Multi-tier pattern)
5. Patterns vs. Tactics

4. Designing Architecture

1. General Design Strategy (Abstraction, Decomposition, Divide & conquer, Generation and test, Iteration, Reuse)
2. Attribute-Driven Design (ADD)
3. Choose a part to design Marshal all ASRs for that part Create and test a design for that part Inputs to and outputs of ADD 8-step process: 1. confirm requirements, 2. choose an element to decompose, 3. identify ASRs, 4. choose a design satisfying ASRs, 5. instantiate elements & allocate responsibilities, 6. define interface, 7. verify & refine requirements, 8. repeat step 2-7 until all ASRs satisfied

5. Documenting Architecture

1. Views and Beyond : Styles (viewpoints), patterns and views Structural views: module views, component-and-connector views, allocation views Quality views
2. Documenting views: 1. build stakeholder/view table, 2. combine views, 3. prioritise & stage
3. Beyond views: documentation info & architecture info (mapping between views)
4. Documentation package: views + beyond

6. Evaluating Architecture

1. Stakeholders involved in ATAM
2. Inputs to and outputs of ATAM
3. Phase 0: Partnership & preparation
4. Phase 1: Evaluation - 1 present ATAM, 2. present business drivers, 3. present architecture, 4. identify architectural approaches, 5. generate utility tree, 6. analyse architectural approaches
5. Phase 2: Evaluation - 2 1. present ATAM & results, 7. brainstorm & prioritize, 8. analyse architectural approaches, 9. present results
6. Phase 3: Follow-up

7. Software Product Lines

1. Software Product Lines (Engineering).

Product = core assets + custom assets Reusability and Modifiability

2. Product Line Architecture

Reuse: find, understand, and use (invoke) Variation: forms of variation * software entity varied * binding time Architecture: variation points

3. SPL Practice Areas and Patterns

29 practice areas and 22 patterns

最终考试

简答题、论述题、设计分析题

英文题目、中文或英文答题

个别题目可能需画图

基础内容70%

高阶内容30%

个人认为设计分析题就是其中的高阶内容。

画图可能还要再准备一下。

往年考题

15年考题

1. Where do software architecture come from? List five possible sources of software architecture.
2. **What distinguishes an architecture for a software product line from an architecture for a simple product?**
3. **How to model quality attribute scenarios? Graphically model two quality attributes in "stimulus-response" format: availability and performance.**
4. Describe relationships between architecture patterns and tactics. List four tactics names and describe their usage.
5. **Briefly describe the general activities involved in a software architecture process.**
6. **Mapping, and list 4 views for each style. (sa07, p.9)**
7. Explain the context, benefits and limitations of Broker Architecture Pattern.
8. Why should a software architecture be documented using different views? Give the name and purposes of 4 example views.
9. Briefly describe the fundamental principles of SOA and discuss the impact of SOA on quality attributes like interoperability, scalability and security.
10. **Describe outputs generated from each phase of ATAM process.**
11. Why SPL and MDA have high reusability? Compare and discuss their commonality and differences.

17年考题

1. **Briefly describe the general activities in a software architecture process, and the major inputs and outputs at each activity.**
2. **What distinguishes an architecture for a software product line from an architecture for a single product?**
3. What are generic design strategies applied in designing software? Give a concise working example with software architecture for each strategy.
4. **How to model quality attribute scenarios? Graphically model two quality attributes in "stimulus-response" format: availability and modifiability.**
5. **Describe outputs generated from each phase of ATAM process.**
6. **Map, and list four views of each category of style.**
7. What are ASR? List four sources and methods for extracting and identifying ASRs.
8. Please name at least three Object-Oriented principles, and explain how they are applied in Strategy pattern?
9. What should be included in a typical software architecture documentation package? Briefly describe each component and its purpose.
10. Describe 4+1 view
11. 软件设计的三个变化维度，每个维度的变化点。differing binding time如何影响可修改性和可测试性。

Variation: forms of variation * software entity varied * binding time

18 年考题 (梁神回忆)

1. 软件架构的关注点有哪些? 利益相关方有哪些?
2. Software requirements, quality attributes, ASRs 的区别和联系
3. What is the nature of component-connector style? 以 MVC pattern 举例
4. 如何对质量属性场景建模? 画出 availability 和 modifiability 的刺激-响应图
5. risks, sensitivity points, trade-off points 是什么? 各举一个例子
6. **连线, 并对每种 style 列出四种视图**
7. Layered pattern 和 Multi-tier pattern 的区别
8. 描述 ADD 过程
9. **为什么软件架构需要用不同的视图描述? 举出四种视图的例子 (列出名称和目的)**
10. 软件产品线架构如何实现可变性? 描述可变性机制的工作方式
11. 设计一个飞行模拟软件, 要求能模拟多种飞机的特性。为了在将来支持更多飞机种类, 要求使用策略模式。画出架构图 and 类图
12. 太复杂, 忘了

19年押题

这 7 个估计考的概率很大, 看明天是不是打脸吧.....

1. ATAM (看规律ATAM 和 ADD 轮流考)
2. 连线, 并对每种 style 列出四种视图 (每年都考, 这个必然了)
3. 各种不同视图 (列举) (4+1 或者后面的)
4. 如何对质量属性场景建模? 画出 availability 和 *** 的刺激-响应图 (可用性是一定要考的, 其他的不好说。概率: 性能 > 可修改性 > 其他)
5. 软件产品线..... (肯定会考, 但是没法猜考什么)
6. 几种设计模式 (很可能考设计题, 要画图)
7. 微服务架构

19 年最终考题 (个人回忆)

1. 如何对质量属性场景建模? 画出 Interoperability 和 modifiability 的刺激-响应图
2. What are ASR? List four sources and methods for extracting and identifying ASRs.
3. 4+1 视图介绍 (还要画图, 我的图画得有点问题, 去wiki百科上可以看)
4. What are generic design strategies applied in designing software? Give a concise working example with software architecture for each strategy. (和17年一样的)
5. Map, and list four views of each category of style. (每年必考题)
6. What should be included in a typical software architecture documentation package? Briefly describe each component and its purpose. (和17年基本一样)
7. 描述 在 ATAM 的每一个过程中 有哪些 stake holder 和他们的职责
8. 软件产品线架构如何实现可变性? 描述可变性机制的工作方式, 和变化点。 (和去年一样)
9. Explain the context, benefits and limitations of Broker Architecture Pattern.
10. 微服务 和 SOA 的区别, 相同点
11. 一个订票系统的设计题, 不同的角色有不同的打折方案, 用策略模式设计, 最后画图, 还要说明策略模式的使用场景。
12. 设计题, 和 15 年的设计题一模一样